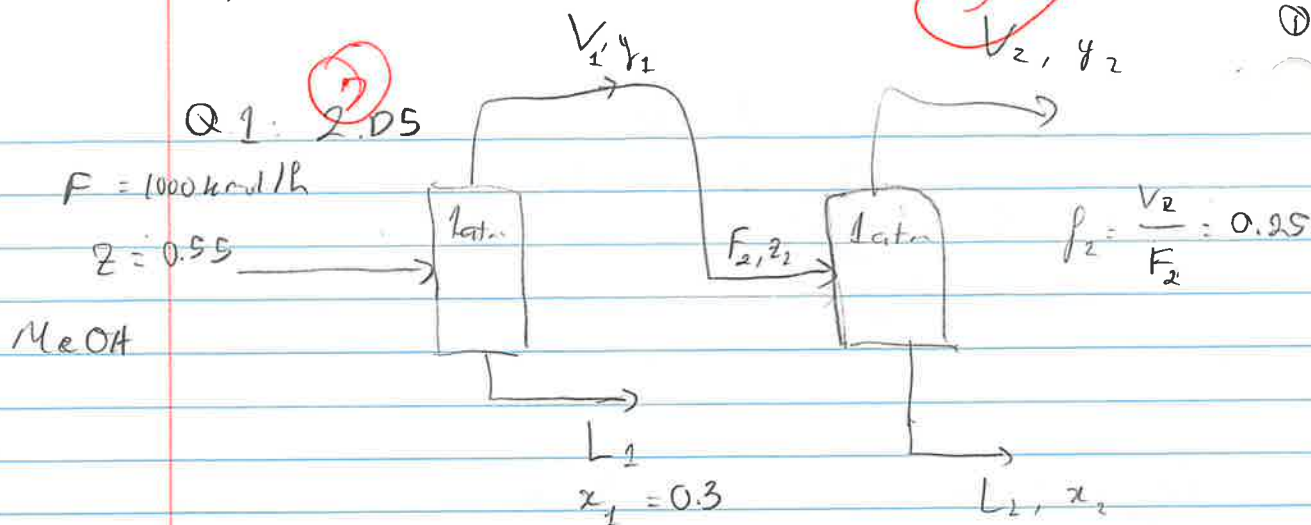




a) From Table 2.8 : $x_1 = 0.3 \Rightarrow y_1 = 0.665$

Balance eqn:

$$\begin{cases} Fz = V_1 y_1 + L_1 x_1 \\ F = V_1 + L_1 \end{cases} \Rightarrow \begin{cases} 1000 \times 0.55 = 0.665 V_1 + 0.3 L_1 \\ 1000 = V_1 + L_1 \end{cases}$$

$$\Rightarrow \begin{cases} L_1 = 315.1 \text{ kmol/h} \\ V_1 = 684.9 \text{ kmol/h} \end{cases}$$

$$\frac{V_1}{F} = \frac{684.9}{1000} = 0.6849$$

From Table 2.8 : $T_1 = 78.0^\circ\text{C}$

b) For simplicity, $F_2 \equiv V_1$ and $z_2 \equiv y_1$

Balance eqns for flash drum 2:

$$\begin{cases} F_2 = V_2 + L_2 \\ F_2 z_2 = L_2 x_2 + V_2 y_2 \end{cases}$$

$$y_2 = \frac{-L_2}{V_2} x_2 + \frac{F_2}{V_2} z_2 = -\frac{F_2 - V_2}{V_2} x_2 + \frac{F_2}{V_2} z_2$$

$$y_2 = -\left(\frac{1-f_2}{f_2}\right) x_2 + \frac{1}{f_2} z_2 \Rightarrow y_2 = -\frac{1-0.25}{0.25} x_2 + \frac{0.665}{0.25}$$

$$y_2 = -3x_2 + \frac{0.665}{0.25}$$

Graphing the above operating equation passing thru $y_1 = x_1, z_1 = 0.665$

From graph, $x_2 = 0.61, y_2 = 0.83$

$T_2 \approx 71.0^\circ\text{C}$

2

Solve system:
$$\begin{cases} 684.9 = L_1 + V_2 \\ 684.9 \cdot 0.665 = 0.61 L_2 + 0.83 V_2 \end{cases}$$

$$\begin{cases} L_2 = 513.7 \text{ kmol/h} \\ V_2 = 171.2 \text{ kmol/h} \end{cases}$$

Q.2. 2.10.10

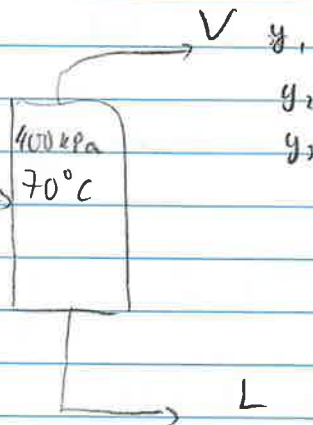
1.6

F

propane: z_1

n-butane: $z_2 = 0.35$

n-hexane: z_3



$T = 70^\circ\text{C} = 617.67^\circ\text{R}$

$P = 400 \text{ kPa} = 58.015 \text{ psia}$

$x_3 = 0.7$

De Priester eqn:

$$\ln k = c \frac{a_{T1}}{T^2} + \frac{a_{T2}}{T} + a_{T6} + a_{p1} \ln p + \frac{a_{p2}}{p^2} + \frac{a_{p3}}{p}$$

Solve for propane:

$$\ln k_1 = \frac{-970688.5625}{617.67^2} + 0 + 7.15059 - 0.76984 \ln(58.015) + 0 + \frac{6.90224}{58.015}$$

$\Rightarrow k_1 = 4.367$

Following similar procedure to find k_2 and k_3 , we have:

$k_2 = 1.967$

$k_3 = 0.923$

$$z_3 = \frac{z_1}{1 + (k_3 - 1)V/F} = 0.7 \Rightarrow z_3 = 0.7 + 0.7(0.323 - 1)V/F$$

$$\Rightarrow z_3 = 0.7 - 0.4739 V/F$$

$$z_1 = 0.65 - z_3 = 0.65 - (0.7 - 0.4739 V/F) = -0.05 + 0.4739 V/F$$

Rachford-Rice eqn:

$$\sum \frac{z_i (k_i - 1)}{1 + (k_i - 1)V/F} = 0 \Rightarrow \frac{z_1 (k_1 - 1)}{1 + (k_1 - 1)V/F} + \frac{z_2 (k_2 - 1)}{1 + (k_2 - 1)V/F} + \frac{z_3 (k_3 - 1)}{1 + (k_3 - 1)V/F} = 0$$

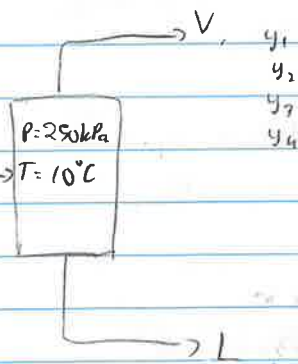
$$\frac{(-0.05 + 0.4739 V/F)(4.387 - 1)}{1 + (4.387 - 1)V/F} + \frac{0.35(1.967 - 1)}{1 + (1.967 - 1)V/F} + \frac{(0.7 - 0.4739 V/F)(0.323 - 1)}{1 + (0.323 - 1)V/F} = 0$$

$$\Rightarrow V/F = 0.581$$

$$\Rightarrow z_3 = 0.7 + 0.7(0.323 - 1)0.581 \Rightarrow z_3 = 0.429$$

Q3: 2.D.16

$F = 150 \text{ kmol/h}$
methane $z_1 = 0.5$
n-butane $z_2 = 0.1$
n-pentane $z_3 = 0.15$
1-hexane $z_4 = 0.25$



Using DePriester eqns for each components:

$$\begin{cases} k_1 = 2.32 \\ k_2 = 0.64 \\ k_3 = 0.18 \\ k_4 = 0.09 \end{cases}$$

④
2

After using Newton approximation on the Rachford-Rice eqn, I found $\frac{V}{F} = 0.240$
(included as Excel attached)

$$V = 0.24 \cdot 150 \Rightarrow V = 36 \text{ kmol/h}$$

$$L = 150 - 36 \Rightarrow L = 114 \text{ kmol/h}$$

Methane balance eqn:

$$\begin{cases} Fz_1 = Lx_1 + Vy_1 \\ y_1 = K_1x_1 \end{cases} \Rightarrow \begin{cases} 150 \cdot 0.5 = 114x_1 + 36y_1 \\ y_1 = 2.32x_1 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = 0.380 \\ y_1 = 0.881 \end{cases} \text{ Methane}$$

n-butane balance eqn:

$$\begin{cases} Fz_2 = Lx_2 + Vy_2 \\ y_2 = K_2x_2 \end{cases} \Rightarrow \begin{cases} 150 \cdot 0.1 = 114x_2 + 36y_2 \\ y_2 = 0.64x_2 \end{cases}$$

$$\Rightarrow \begin{cases} x_2 = 0.109 \\ y_2 = 0.070 \end{cases} \text{ n-butane}$$

n-pentane balance eqn:

$$\begin{cases} Fz_3 = Lx_3 + Vy_3 \\ y_3 = K_3x_3 \end{cases} \Rightarrow \begin{cases} 150 \cdot 0.15 = 114x_3 + 36y_3 \\ y_3 = 0.18x_3 \end{cases}$$

$$\Rightarrow \begin{cases} x_3 = 0.187 \\ y_3 = 0.033 \end{cases} \text{ n-pentane}$$

n-hexane balance eqn:

$$\begin{cases} Fz_4 = Lx_4 + Vy_4 \\ y_4 = K_4x_4 \end{cases} \Rightarrow \begin{cases} 150 \cdot 0.25 = 114x_4 + 36y_4 \\ y_4 = 0.05x_4 \end{cases}$$

$$\Rightarrow \begin{cases} x_4 = 0.323 \\ y_4 = 0.017 \end{cases} \text{ n-hexane}$$